

Biodiversity

Biodiversity conservation methods

There are many factors that underlie the need to conserve biodiversity, such as,

- present and potential uses of the components of biological diversity - especially as we have no way of knowing or predicting what will be of use in the future.
- biodiversity is essential to maintain the earth's life support systems that enable the biosphere to support human life.
- It is ethically important to maintain all of the earth's biological diversity, including all the other extant (currently existing) life forms.

Biodiversity conservation:

Biodiversity conservation refers to the protection, upliftment, and management of biodiversity in order to derive sustainable benefits for present and future generations. A wide variety of species will cope better with threats than a limited number of them in large populations. Even if certain species are affected by pollution, climate change or human activities, the ecosystem as a whole may adapt and survive.

It is the practice of **protecting** and **preserving** the wealth and variety of species, habitats, ecosystems and the genetic diversity on the planet.

- In addition to protection of resources, it is also the **rational use** of natural resources.
- It is essential for our health, wealth, food, fuel and services we depend on
- It also plays an important role in supporting several sectors of development.



Fig: - Biodiversity

Aim of conservation?

1. **Minimize** depletion of **resources**.
2. **Preserve resources** for use by future generations.

Approach for biodiversity conservation:

1. **In-situ conservation**
2. **Ex-situ conservation**

Ways of conservation:

- **By law:** giving protection to animals and plants or special areas of land or water

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- **Restoration** of unattractive countryside like waste spills, slag heaps, etc.
- **Rewilding:** allowing areas to restore themselves naturally without human interference.
- **Alternative energy:** a need to find alternative resources to replace coal, oil, etc.
- **Nature reserves & Zoos.**
- **Recycling:** Reusing of unwanted products such as newspapers, scrap metal, glass, sewage, etc.
- **Education and awareness:** Spreading the message of conservation to schools, youth organizations and the media to achieve maximum results.



1. In-situ conservation:

In-situ conservation means “on-site conservation”. It is the process of protecting an endangered plant or animal species in its natural habitat, either by protecting or cleaning up the habitat itself, or by defending the species from predators. The benefit to in-situ conservation is that it maintains recovering populations in the surroundings where they have developed their distinctive properties. Wildlife conservation is mostly based on in-situ conservation. This involves the protection of wildlife habitats. Also, sufficiently large reserves are maintained to enable the target species to exist in large numbers.

- In situ conservation is the preservation of species and populations of living organisms in a natural state in the habitat where they naturally occur.
- This can be achieved either by:
 - Protecting or cleaning up the habitat itself.
 - Defending the species from predators.

Methods of in-situ conservation:

1. Biosphere reserves
2. National parks
3. Wildlife sanctuaries
4. Tiger reserves
5. Gene sanctuaries
6. Community reserves
7. Sacred groves

1. Biosphere reserves:

Biosphere reserves cover very large areas, often **more than 5000 km²**. They are used to protect species for a long time. Currently, there are **18 in India**.

Example: Manas (Assam), Gulf of Mannar (Tamil Nadu), Nanda Devi (U.P)

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2. National parks:

A national park is an area dedicated for the **conservation of wildlife along with its environment**, including its scenery, natural and historical objects. It ranges from **100-500 km²**. Within biosphere reserves, one or more national parks may also exist. Currently, there are **106 national parks** in India.

Example: Gir National Park (Gujarat), Bandipur (Karnataka), Periyar (Keral)

National parks are largely natural and unchanged by human activities, but many of them already had existing human impacts before they were designated for protection and human activities have often been allowed to continue. People have no rights in a National Park.

3. Wildlife sanctuaries:

A wildlife sanctuary is an area which is **reserved for the conservation of animals only**. Currently, there are **551 wildlife sanctuaries** in India. The first wildlife sanctuary was the Vedanthangal Bird Sanctuary near Madras, set up in 1878, which merely formalised the traditional protection afforded by villagers for pelicans, herons and other birds breeding at Vedanthangal. Another such sanctuary was set up at Ranganathittu near Mysore, in 1942.

Reserves for specific animals:

Several projects have been setup in our country for the protection of specific animals

Example: Project Tiger, Gir Lion Project, Crocodile Breeding Project, Project Elephant, Snow Leopard Project, etc.

4. Tiger reserves:

Project Tiger was launched in 1973 to save the tiger. Starting from 9 reserves in 1973, it has grown to 29 in 2006 covering a **geographical area of 1.17%**

E. g.: Periyar, Kanha, Corbett

5. Gene sanctuary:

Gene sanctuary is an area where plants are conserved, including both biosphere reserves and national parks. The first gene sanctuary in India has been setup in the Garo hills, Meghalaya for wild relatives of citrus.

6. Community reserves:

It is a kind of protected area to provide legal support to community/privately owned reserves that cannot be designated as national parks or wildlife sanctuaries. There are 218 existing

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Community Reserves in India covering an area of 1445 km², which is 0.044% of the geographical area of the country. (National Wildlife Database, Dec. 2021). Keshopur chamb gurdaspur (Punjab) conservation reserve India's first community reserve.

7. Sacred groves:

They are areas of forests set aside, usually for tribal communities, where all trees and wildlife within are venerated and given total protection. The examples of sacred groves are Khasi and Jaintia Hills in Meghalaya and Aravalli Hills of Rajasthan. - India has a history of religious and cultural traditions that emphasized the protection of nature.

2. Ex-situ conservation:

Ex-situ conservation means, literally “off-site conservation”. It is the process of protecting population of an endangered species of plant or animal by removing it from an unsafe or threatened habitat and placing it, or part of it, under the care of humans. While ex-situ conservation is comprised of some of the oldest and best-known conservation methods known to human, it also involves newer, sometimes controversial laboratory methods. Ex-situ conservation, while helpful in human’s efforts to sustain and protect our environment, is rarely enough to save a species from extinction. It is to be used as a last resort or as a supplement to in-situ conservation because it cannot recreate the habitat as a whole: the entire genetic variation of a species, its symbiotic counterparts, or those elements which, over time, might help a species adapt to its changing surroundings. Furthermore, ex-situ conservation techniques are often costly. Plants and animals living in ex-situ breeding grounds have no natural defense to the diseases and pests new to the species.

This is usually done by **removing a part of the population** from a threatened habitat and placing it in a new location.

Different ex-situ conservation methods:

1. Botanical gardens
2. Zoos
3. Seed banking
4. Cryopreservation
5. Herbal gardens
6. Plant herbariums

1. Botanical gardens:

They are one of the most conventional methods of ex-situ conservation of plants. India has more than **100 botanical gardens** under different management systems located in different bio-geographical regions. Globally, there are around 2000 botanical gardens in ~148 countries. Central and state governments manage 33 botanical gardens that maintain the diversity in the form of plants or plant populations. These facilities provide not only **housing and care** for

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specimens of endangered species, but also have an **educational value**, informing the public of the threatened status of endangered species.

Example: Hyderabad Botanical Garden (Telangana), Panjab University Botanical Garden (Chandigarh).

2. Zoos:

Zoos are some of most publicly visited ex situ conservation sites, with the WZCS (World Zoo Conservation Strategy) estimating that the 1100 organized zoos in the world receive more than 600 million visitors annually. It has been estimated to be a total of **2,107 aquaria and zoos in 125 countries**, in addition to privately owned facilities. According to the Zoo Authority of India, there are **~164 zoos in India**. Example: national zoological park (Delhi), Rajiv Gandhi Zoological Park (Pune).

3. Seed banking:

A seed bank stores seeds to preserve genetic diversity. One of the most efficient methods of ex-situ conservation for sexually reproducing plants is the storage of conservation material in form of seeds. In this process we need to store the seeds in a temperature and moisture-controlled environment.

4. Cryopreservation:

Cryopreservation is the only ex situ conservation method for long-term preservation of species that cannot be stored in seed banks. Plant cryopreservation consist of the storage of seeds, pollen, tissue, or embryos in liquid nitrogen.

5. Herbal gardens:

Herbal gardens refer to gardens that conserve herbs, shrubs that are of **medicinal value and aromatic value**.

6. Plant herbariums:

- Herbariums preserve plant diversity for research and breeding purposes, often acting as dictionaries of plant kingdoms
- The Botanical Survey of India has the largest holding of 1,500,000 specimens.
 - E.g.: Presidency College Madras (1,00,000)
 - St. Joseph's College, Tiruchirapally (60,000)
